Windows Troubleshooting

Student lab guide · IT Support · complete Part 0 **before** class

How to use this guide

Part 0 is setup you do on your own time. It takes 45 to 90 minutes, most of which is unattended download and install, so start it the evening before. Parts 1 to 5 are the labs we run together in class. Part 6 is practice for afterwards.

If you run out of time on Part 0, still come to class. There is a no-install fallback in the box below and you will not be left behind.

Part 0 · Set up a Windows lab machine

Never practise troubleshooting on the computer you depend on. A virtual machine is a full Windows install running in a window on your real desktop. If you break it, you delete it and make another. That freedom is the whole point.

Which path is yours

Your computer	Do this
Windows 10 or 11, and it is *your* machine	You may use it directly for Parts 1 to 4. Do Part 5 (Restore) in a VM only.
Windows, but it is a work or shared machine	Use VirtualBox. Do not run diagnostics on hardware you do not own.
Mac with Apple Silicon (M1/M2/M3/M4)	Use UTM. VirtualBox does not run Windows on Apple Silicon.
Mac with an Intel chip	Use VirtualBox, same as Windows users.
Chromebook, or a machine under 8 GB RAM	Use the no-install fallback below. Tell your instructor before class.

Option A · VirtualBox (Windows and Intel Macs)

1. Download and install **Oracle VirtualBox** from virtualbox.org. Accept the network driver prompt.
2. Download the **Windows 11 Disk Image (ISO)** from microsoft.com/software-download/windows11. It is about 5 GB. Start this first and let it run while you do other things.
3. In VirtualBox click **New**. Name it *WinLab*, type Microsoft Windows, version Windows 11 64-bit.
4. Select your ISO in the same dialog and tick **Skip Unattended Installation**.
5. Give it **4096 MB RAM** and **2 CPUs**. If your machine has 16 GB or more, use 8192 MB.

6. Create a virtual hard disk of **60 GB**, dynamically allocated.
7. In Settings > System > Motherboard, confirm **Enable EFI** is ticked. Windows 11 will not install without it.
8. Start the VM and follow the Windows installer. Choose **I don't have a product key** — the evaluation copy is fine for coursework.
9. When Windows asks you to sign in, look for **Sign-in options** and choose a local account so you do not tie this throwaway machine to your real Microsoft account.
10. Once the desktop loads, install **Guest Additions** from the Devices menu, then restart.

If the VM will not start: virtualisation is almost certainly disabled in your BIOS. Restart, enter BIOS setup (usually F2, F10 or Del), and enable Intel VT-x or AMD-V. This is the single most common failure and it is a two-minute fix.

Option B · UTM (Apple Silicon Macs)

1. Install **UTM** from mac.getutm.app. The direct download is free; the App Store version is a paid convenience.
2. In UTM choose **Create a New Virtual Machine** > **Virtualize** > **Windows**.
3. Tick **Import VHDX Image** is *off*, then tick **Fetch latest Windows installer**. UTM downloads the ARM build of Windows 11 for you, which is the build Apple Silicon needs.
4. Allocate **4 GB RAM** and **64 GB storage**.
5. Complete the Windows installer as above, choosing a local account.
6. Install **SPICE Guest Tools** when prompted so that display resizing and clipboard sharing work.

Apple Silicon runs the ARM version of Windows. Everything in this lab works identically. A few third-party x86 tools will not install, which is worth knowing but does not affect us today.

Fallback · no install required

If none of the above is possible, you can still do every lab except Part 5. Come to class and say so in chat at the start. Your instructor will pair you with someone who has a working VM, and you will drive while they share their screen. Watching someone else type is worth far less than typing yourself, so treat this as temporary and get your VM running before the next session.

Before you close the laptop

- Take a **snapshot** in VirtualBox (Machine > Take Snapshot) or UTM. Name it *clean*.
- This is your undo button for the entire course. If a lab wrecks the VM, roll back to *clean* in seconds.
- Snapshot before every risky lab. Professionals do this. It is not a beginner crutch.

In-class labs

We run these together. Tick each one off as you finish and report in chat.

Part 1 · Task Manager

1. Press **Ctrl + Shift + Esc**.
2. If you see a small window, click **More details**.
3. Open the **Processes** tab. Click the **Memory** column heading to sort highest first.
4. Note the top process. Put its name in chat.
5. Click the **Startup apps** tab. Count how many are set to Enabled.

Think about it: Startup apps run every time the machine boots, before the user does anything. A machine with twenty enabled startup apps is slow by design, not by fault. This is the fix for a large share of 'my computer got slow' tickets.

Part 2 · Built-in troubleshooters

1. Open **Settings > System > Troubleshoot > Other troubleshooters**. On Windows 10 it is Settings > Update & Security > Troubleshoot.
2. Find **Playing Audio** and click Run.
3. Follow it all the way to the end. Do not close it early.
4. Report in chat what it found, **even if it found nothing**. A clean result is still a result.

Connect it to today: if you had to pick a different speaker or microphone to join this call, you did manually what this tool automates. That is not a coincidence — wrong default audio device is the most common audio ticket there is.

Part 3 · Find Safe Mode (do not boot into it)

1. Open **Settings > System > Recovery**.
2. Find **Advanced startup** and its **Restart now** button.
3. **Stop there.** Do not click it during class or you will drop off the call.
4. Put a hand-raise or ✓ in chat once you can see the button.

The full path once you do restart is: **Troubleshoot > Advanced options > Startup Settings > Restart**, then press **4** for Safe Mode or **5** for Safe Mode with Networking. On a machine that will not boot at all, interrupting startup three times sends Windows into this same recovery environment automatically.

Part 4 · Disk health

Open an **administrator** Command Prompt: right-click Start, choose Terminal (Admin) or Command Prompt (Admin).

```
chkdsk C:
```

Read-only. Safe to run any time. Reports problems without changing anything. Run this one now and paste the last line of the output into chat.

```
chkdsk C: /f /r
```

Fixes errors and attempts to recover readable information. Requires exclusive access to the drive, so it schedules itself for the next restart. Can take several hours on a large disk. **Do not run this during class.**

```
wmic diskdrive get model,status
```

A quick S.M.A.R.T. summary. Returns **OK** or **Pred Fail**. For real detail, CrystalDiskInfo shows reallocated sector count, power-on hours and temperature.

The distinction that gets asked in interviews: chkdsk checks the *filesystem* — whether the index of where files live is consistent. S.M.A.R.T. checks the *physical device* — whether the hardware is wearing out. A drive can pass one and fail the other. Rising reallocated sectors means back up now, not repair.

Part 5 · System Restore (VM only)

1. Press **Windows + R**, type **rstrui.exe**, press Enter.
2. Look at the list of available restore points. Note the date of the oldest.
3. If the list is empty, protection was never enabled. Close this and go to **System Properties > System Protection > Configure** to turn it on.
4. **In your VM only:** take a snapshot first, then actually run a restore and watch what happens.

	System Restore	System Reset
Changes	System files, registry, drivers, programs	Reinstalls Windows entirely
Personal files	Untouched	Kept or removed — your choice, and it is final
Installed apps	Anything added after the restore point is gone	All removed
Typical time	15 to 30 minutes	An hour or more
Use when	It broke recently and you know roughly when	Corruption, malware, or nothing else worked

Back up before either one. Restore can fail. Reset is irreversible. A technician who resets a machine without checking where the user's files live does more damage than the original fault.

Part 6 · Practice on your own

- Snapshot your VM, then deliberately break something: disable a network adapter in Device Manager, or set a wrong default audio device. Now fix it using only the tools from today.
- Write a three-sentence ticket note for the fix: what the symptom was, what you did, what to watch for.

- Roll back to your *clean* snapshot and do it again with a different fault.

Breaking things on purpose in a safe environment is the fastest way to learn this job. It is also the only way to build the confidence to touch a stranger's machine.